

Pattern Recognition Letters

Genesis: Evolving Optimal Cache Schedule for Diffusion Models

--Manuscript Draft--

Manuscript Number:	PRLETTERS-D-26-01556
Article Type:	Regular Article
Keywords:	Evolutionary Algorithms, Island Model, Optimization, Caching, Stable Diffusion
Corresponding Author:	Deepanshi Sharma Indian Institute of Technology Mandi
First Author:	Deepanshi Sharma
Order of Authors:	Deepanshi Sharma Parimala Kancharla
Abstract:	Diffusion models have revolutionized image generation by producing very high quality results but their deployment remains constrained by slow inference due to sequential denoising steps. Training free acceleration methods like DeepCache exploit feature redundancy across timesteps where determining an optimal caching schedule still remains a challenging task. In this paper, we propose Genesis which is an evolutionary optimization framework that automatically discovers optimal caching schedule using Island Model. We also introduce diffusion aware evolutionary operators Phase Aware Crossover and Block Mutation. These operators preserve temporal coherence across denoising steps and enable a more efficient exploration of the search space. We also provide a rigorous evaluation protocol including runtime and memory profiling along with task specific Pareto frontier analysis to characterize the trade off between inference efficiency and image quality. Experiments on MS-COCO and PartiPrompt datasets shows that our method achieves a 60.1% reduction in BOPs and quality trade off as compared to DeepCache, reduces computational cost, maintains high quality and prompt alignment as measured by FID and CLIPScore.

Pattern Recognition Letters

Authorship Confirmation

As corresponding author I, Deepanshi Sharma, hereby confirm on behalf of all authors that:

1. This manuscript or a large part of it has not been published and is not being submitted to any other journal.
2. All text and graphics except for those marked with sources, are original works of the authors and all necessary permissions for publication were secured prior to submission of the manuscript.
3. All authors each made a significant contribution to the research reported and have read and approved the submitted manuscript.

Signature: Deepanshi Sharma

Date: 11 July, 2026

List any pre-prints:

Relevant Conference publication(s) (submitted, accepted, or published):

Justification for re-publication:

Pattern Recognition Letters

Genesis: Evolving Optimal Cache Schedule for Diffusion Models

Dear Editor,

Thank you for your useful comments and suggestions on the language and structure of our manuscript. We have modified the manuscript accordingly, and detailed corrections are listed below point by point:

1. Each research highlight provided in bullet points should not exceed a maximum of 85 characters (including spaces) per bullet point.

✓ We have revised the highlights, now all the bullet points are less than 85 characters (including spaces).

The manuscript has been resubmitted to your journal. We look forward to your positive response.

Sincerely,

Deepanshi Sharma

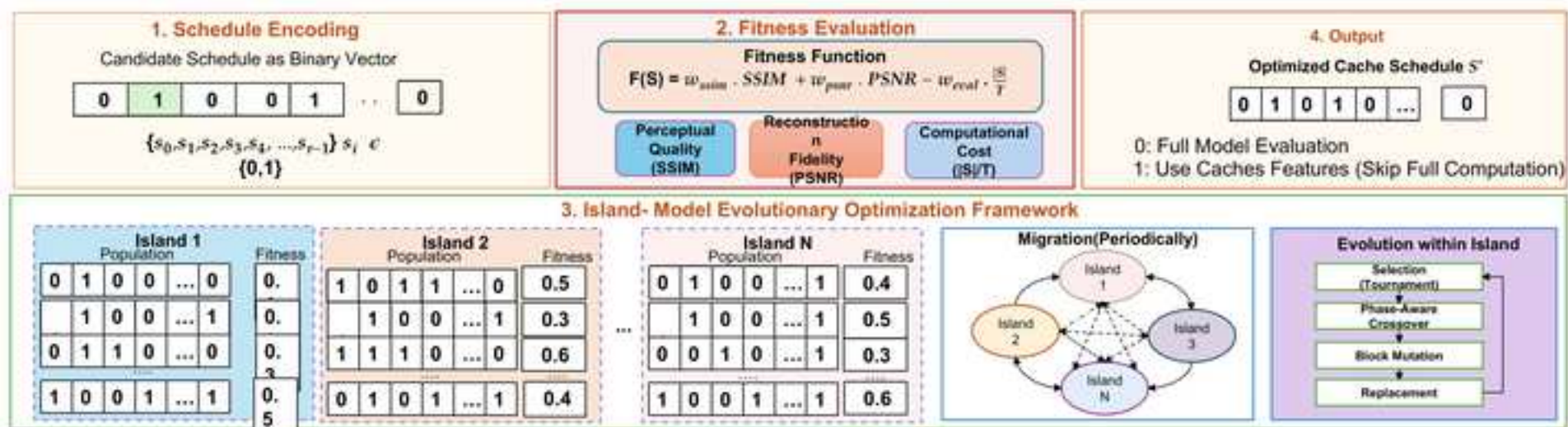


Figure 1. Architecture for Genesis

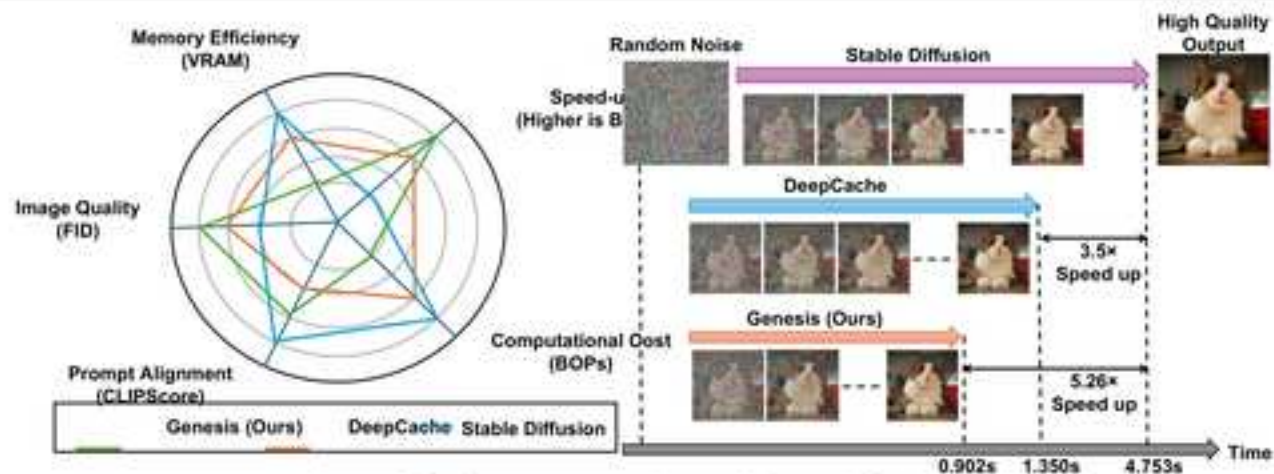


Figure 2. Performance comparison of Proposed Methodology

Genesis: Evolving Optimal Cache Schedule for Diffusion Models

Deepanshi Sharma^{a,*} (Research Scholar), Dr. Parimala Kancharla^a (Assistant Professor)

^a*School of Computing and Electrical Engineering, Indian Institute of Technology Mandi, 175075, Himachal Pradesh, India*

ARTICLE INFO

Keywords:

Evolutionary Algorithms
Island Model
Optimization
Caching
Stable Diffusion.

ABSTRACT

Diffusion models have revolutionized image generation by producing very high quality results but their deployment remains constrained by slow inference due to sequential denoising steps. Training free acceleration methods like DeepCache exploit feature redundancy across timesteps where determining an optimal caching schedule still remains a challenging task. In this paper, we propose Genesis which is an evolutionary optimization framework that automatically discovers optimal caching schedule using Island Model. We also introduce diffusion aware evolutionary operators Phase Aware Crossover and Block Mutation. These operators preserve temporal coherence across denoising steps and enable a more efficient exploration of the search space. We also provide a rigorous evaluation protocol including runtime and memory profiling along with task specific Pareto frontier analysis to characterize the trade off between inference efficiency and image quality. Experiments on MS-COCO and PartiPrompt datasets shows that our method achieves a 60.1% reduction in BOPs and quality trade off as compared to DeepCache, reduces computational cost, maintains high quality and prompt alignment as measured by FID and CLIPScore. The code is available at <https://github.com/Deep712sharma/Genesis.git>

1. Introduction

Diffusion models[Rombach, Blattmann, Lorenz, Esser and Ommer (2022), Weng (2021)] have transformed the field of generation. Unlike earlier approaches like GANs[Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville and Bengio (2020)], which generate output in a single step, diffusion models work by gradually refining noise into meaningful data over multiple timesteps. This process leads to better image quality and high computational cost.

To reduce this computational burden, recent research has focused on minimizing the number of inference steps. One of the promising training free approaches is DeepCache[Ma, Fang and Wang (2024)]. It is based on the observation that high level semantic features in U-Net[Ibtehaz and Kihara (2023)] change very little between consecutive denoising steps. By caching these intermediate features and reusing them avoids redundant computation and speed up inference without retraining or fine tuning.

One of the key challenges is determining the optimal cache schedule. This schedule defines when the full model should be executed and when the cached features should be used. The search space is large, discrete and highly irregular. So finding the optimal cache schedule is a problem.

In this paper, we address this challenge by framing schedule discovery as an evolutionary[Zhang, Hartl, Hazan and Levin (2024)] optimization problem. We introduce an Island model[Whitley, Rana and Heckendorn (1999)] based searching algorithm specifically designed to explore the complex search space of caching schedules. Diffusion models exhibit strong temporal correlations across denoising steps where early steps determine global structure and later steps refine fine details. To use this property we design diffusion specific

evolutionary operators that preserve phase consistency and exploit local smoothness in search space which enables efficient and meaningful exploration.

To validate effectiveness of our method, we tested it on well known optimization benchmarks such as Rosenbrock, Ackley and Rastrigin functions and evaluated on COCO[Lin, Maire, Belongie, Hays, Perona, Ramanan, Dollár and Zitnick (2014)] and PartiPrompt[Yu, Xu, Koh, Luong, Baid, Wang, Vasudevan, Ku, Yang, Ayan et al. (2022)] datasets for text to image generation which shows our approach successfully identifies the cache schedule that offer faster inference and high image quality along with prompt alignment. Figure 1 illustrates the overall motivation of the proposed Genesis framework by comparing it with Stable Diffusion and DeepCache in terms of inference speed, computational efficiency, memory usage, and image quality. In summary, our main contributions are as follows:

- We reformulate DeepCache scheduling as a structured evolutionary optimization problem.
- We introduce diffusion aware operators Phase Aware Crossover and Block Mutation which exploit temporal structure in diffusion process.
- We provide an experimental evaluation of runtime and VRAM profiling along with task specific Pareto frontier analysis to maintain speed-quality trade off.
- We demonstrate in Stable Diffusion using MS-COCO and PartiPrompt datasets that our optimized schedule achieves better efficiency & high image quality.

2. Related Works

Diffusion models[Weng (2021), Rombach et al. (2022)] achieve high-quality generative performance but require iterative denoising steps, making inference slower and more resource intensive. Early acceleration strategies focused on reducing the number of sampling steps using improved solvers such as DDIM, DPM-Solver. More recent approaches explore architectural modifications or knowledge distillation to smaller student models; however, these methods require

*Corresponding author.

✉ s24135@students.iitmandi.ac.in (D. Sharma);

parimala@iitmandi.ac.in (Dr.P. Kancharla)

ORCID(S): 0009-0002-3094-5761 (D. Sharma); 0009-0001-4113-1600 (Dr.P. Kancharla)

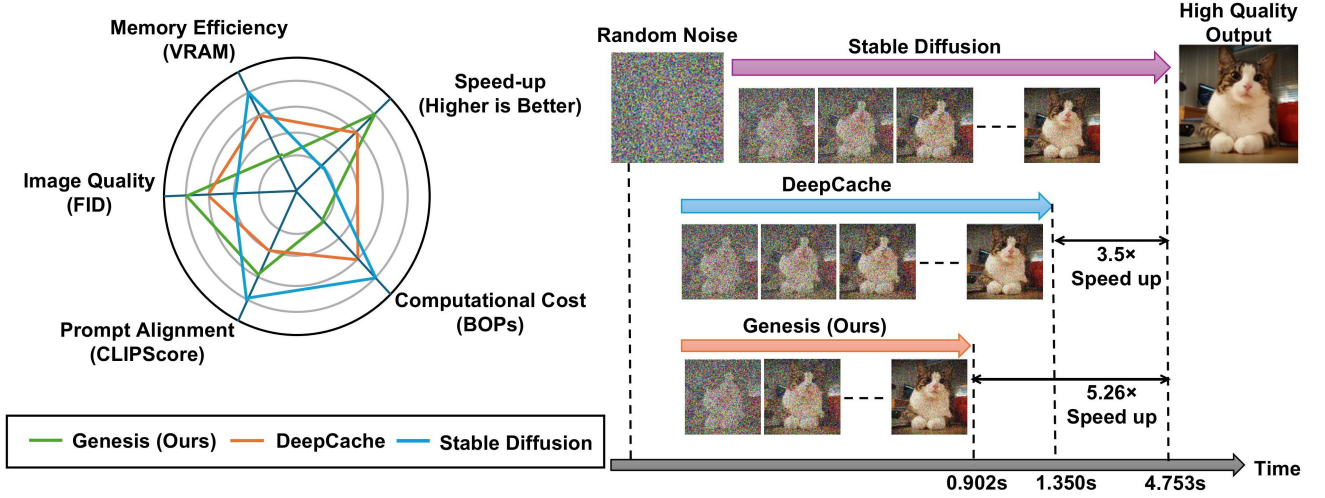


Figure 1: Comparison of Genesis, DeepCache, and Stable Diffusion. Genesis provides better overall performance across multiple metrics and reduces inference time through an optimized cache schedule.

retraining and may not generalize across diverse diffusion backbones.

Diffusion models[Weng (2021), Rombach et al. (2022)] process highly similar latent representations across timesteps suggesting opportunities for activation caching. Early caching approaches like DeepCache[Ma et al. (2024)], DreamCache and Learning to Cache reused attention maps or intermediate feature blocks to reduce redundant compute. More recent works like CacheQuant[Liu, Li and Gu (2025)] and Q&C explicitly formalize activation reuse for diffusion models selecting specific blocks or timesteps where cached activations can be substituted with minimal error.

Evolutionary[Zhang et al. (2024)] computation has become an effective solution for solving difficult optimization problems characterized by discontinuous, multimodal, or combinatorial search spaces. Population-based methods such as Genetic Algorithms[Whitley et al. (1999)] (GAs) perform global optimization through iterative selection, crossover, and mutation, requiring only fitness evaluations rather than gradient information. Their flexibility has led to applications in scheduling, feature selection, architecture search and resource allocation. Several studies have demonstrated that Island Model Genetic Algorithms outperform standard GAs by evolving multiple subpopulations independently while periodically exchanging elite individuals. This migration strategy preserves genetic diversity, mitigates premature convergence, and improves scalability on large optimization problems. Although evolutionary methods have recently been investigated in the context of generative modeling, they have rarely been explored for optimizing activation reuse or cache scheduling in diffusion models. Our approach addresses this gap by introducing an island-based evolutionary framework together with diffusion-aware crossover and mutation operators that exploit the temporal structure of denoising trajectories.

3. Methodology

In this paper, we propose an evolutionary[Ao (2005)Baldwin (2018)] optimization framework to improve the caching strategy in diffusion models. Instead of relying on fixed and

uniform caching intervals our approach searches for adaptive schedules that determine when to perform full computation and when to reuse cached features. To achieve this we used an Island Model. Complete pipeline is described in figure 2.

Algorithm 1:

Input: $T, N, G_{max}, M_{int}, \mathbf{I}_{base}$
Output: $\mathbf{S}^*$

```

 $\mathbf{S}^* \leftarrow \emptyset, \mathcal{F}^* \leftarrow -\infty;$ 
 $\{\mathcal{P}_1, \dots, \mathcal{P}_N\} \leftarrow \text{InitializePopulation}();$ 
for  $g \leftarrow 1$  to  $G_{max}$  do
    for  $n \leftarrow 1$  to  $N$  do
        foreach  $\mathbf{S} \in \mathcal{P}_n$  do
             $\mathbf{I}_{dist} \leftarrow \text{Generate}(\mathbf{S});$ 
             $\mathcal{F}(\mathbf{S}) \leftarrow \text{Fitness}(\mathbf{I}_{base}, \mathbf{I}_{dist}, |\mathbf{S}|);$ 
            if  $\mathcal{F}(\mathbf{S}) > \mathcal{F}^*$  then
                 $\mathbf{S}^* \leftarrow \mathbf{S};$ 
                 $\mathcal{F}^* \leftarrow \mathcal{F}(\mathbf{S});$ 
         $\mathcal{P}_n^e \leftarrow \text{Selection}(\mathcal{P}_n);$ 
         $\mathcal{P}_n^c \leftarrow \text{PhaseCrossover}(\mathcal{P}_n^e);$ 
         $\mathcal{P}_n^m \leftarrow \text{BlockMutate}(\mathcal{P}_n^c);$ 
         $\mathcal{P}_n \leftarrow \mathcal{P}_n^e \cup \mathcal{P}_n^m;$ 
    if  $g \bmod M_{int} = 0$  then
        for  $n \leftarrow 1$  to  $N$  do
             $\mathcal{E}_n \leftarrow \text{Top}(\mathcal{P}_n, k);$ 
             $tgt \leftarrow (n \bmod N) + 1;$ 
             $\mathcal{P}_{tgt} \leftarrow \text{ReplaceWorst}(\mathcal{P}_{tgt}, \mathcal{E}_n);$ 
return  $\mathbf{S}^*;$ 
    
```

3.1. Problem Formulation

Diffusion models generate images through a sequential denoising process over T timesteps. At each timestep, the model refines the image progressively. We define caching schedule as:

$$\mathbf{S} = [s_0, s_1, \dots, s_k], \quad s_i \in \{0, 1, \dots, T-1\} \quad (1)$$

where s_i represents timesteps in which the full evaluation is performed. All remaining steps use cached features. The

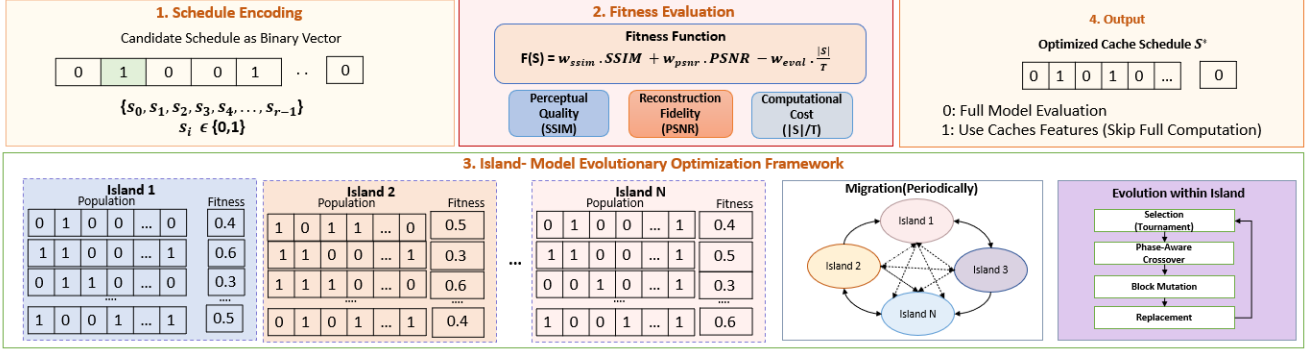


Figure 2: Overview of the proposed evolutionary caching framework for diffusion models. The caching mechanism selectively reuses backbone and skip features to reduce computation. An island-model optimizes the caching schedule S (set of reused timesteps), where multiple islands evolve candidate solutions in parallel and periodically exchange information. The fitness function $F(S)$ balances perceptual quality (SSIM), reconstruction fidelity (PSNR)[Tanchenko (2014)], and computational cost, guiding the search toward an optimal cache schedule that minimizes inference overhead while preserving image quality.

initial timestep is always evaluated as $s_0 = 0$. Our objective is to find the optimal schedule:

$$S^* = \arg \max_S F(S) \quad (2)$$

3.2. Background

Diffusion models[Rombach et al. (2022)] are widely known for producing high quality images. At higher level they work by gradually corrupting data and then learning how to reverse that corruption. In the forward process noise is added step by step to an image:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I) \quad (3)$$

where x_t is the noisy image at timestep t and β_t controls how much noise is added. After many steps the image becomes pure Gaussian noise. In reverse process, a neural network learns to remove noise:

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I) \quad (4)$$

Caching[Ma et al. (2024)] avoids recomputing expensive operations by reusing previously computed results. In diffusion models let the U-Net be decomposed into:

$$f(x_t) = f_{\text{deep}}(x_t) + f_{\text{shallow}}(x_t) \quad (5)$$

Empirically deeper features change slowly:

$$f_{\text{deep}}(x_t) \approx f_{\text{deep}}(x_{t+1}) \quad (6)$$

So these similar features can be cached and reused later.

Evolutionary algorithms[Zhang et al. (2024)] optimize problems by simulating a natural selection process. We define a population:

$$P = \{S_1, S_2, \dots, S_n\} \quad (7)$$

where S_i is a candidate solution or an individual. Each individual is evaluated using a fitness function:

$$F(S) = \alpha \cdot \text{Quality}(S) - \beta \cdot \text{Cost}(S) \quad (8)$$

The probability of selection of an individual is:

$$P(S_i) = \frac{F(S_i)}{\sum_j F(S_j)} \quad (9)$$

New solutions or offsprings[Darwin (2016)Gardner and Welch (2011)] are generated by Crossover and Mutation.

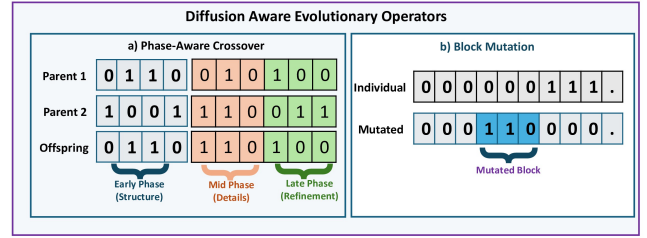


Figure 3: Proposed Evolutionary Operators.

3.3. Island Model

Due to combinatorial and discontinuous nature of search space we adopt an Island Model to improve exploration and prevent premature convergence.

3.3.1. Encoding and Population Structure

Each candidate schedule is encoded as a binary vector:

$$\mathbf{x} = [x_0, x_1, \dots, x_{T-1}], \quad x_t \in \{0, 1\} \quad (10)$$

where, $x_t = 1$ denotes full evaluation and $x_t = 0$ denotes cached reuse. $x_0 = 1$ ensures that the first timestep is always evaluated. The global population is divided into N islands. Each island evolves independently to maintain diversity.

3.3.2. Proposed Evolutionary Operators

Phase Aware Crossover Diffusion models have different denoising phases. Early timesteps determine global structure and later timesteps refine details. Random mixing of these phases disrupts temporal consistency. To address this, we propose Phase Aware Crossover which partitions the timestep sequence into phases:

$$\mathbf{x} = [\mathbf{x}^{\text{early}}, \mathbf{x}^{\text{mid}}, \mathbf{x}^{\text{late}}] \quad (11)$$

Crossover is performed at phase boundaries:

$$\mathbf{x}^{\text{child}} = [\mathbf{x}_{(1)}^{\text{early}}, \mathbf{x}_{(2)}^{\text{mid}}, \mathbf{x}_{(1)}^{\text{late}}] \quad (12)$$

where parents (1) and (2) contribute entire phases instead of randomly mixing individual bits as shown in fig 3. This preserves semantic consistency across denoising steps.

Block Mutation Bit flip mutation ignores the correlation between adjacent timesteps in diffusion models. Since adjacent steps shows high feature similarity we propose Block Mutation which flips contiguous segments:

Table 1

Comparison on MS-COCO and PartiPrompts[Yu et al. (2022)] for text to image generation. FID was computed using 5,000 samples. The BOPs for CacheQuant is exceptionally less because it is using both caching and quantization but all other methods are using caching only.

Method	BOPs↓	MS-COCO (5000 samples)				PartiPrompts[Yu et al. (2022)]		
		FID↓	CLIP↑	PSNR↑	SSIM↑	CLIP↑	PSNR↑	SSIM↑
PLMS-50 steps	346.96T	25.59	26.89	—	1.000	27.23	—	1.000
PLMS-20 steps	138.78T	24.70	26.74	32.14	0.845	27.04	30.50	0.820
TGATE	130.2T	24.12	26.70	31.85	0.832	26.80	30.25	0.815
Δ-DiT	125.4T	23.95	26.75	32.05	0.840	26.91	30.40	0.818
TokenCache	128.6T	23.80	26.72	31.90	0.835	26.85	30.30	0.816
CacheQuant	4.27T	23.55	26.70	32.10	0.842	26.82	30.45	0.819
DeepCache[Ma et al. (2024)]	133.58T	23.45	26.71	32.45	0.851	26.86	31.10	0.830
Ours (Genesis)	80.00T	20.32	26.83	34.12	0.895	27.12	32.85	0.855

Table 2

Ablation study isolating the contributions of the Island Model and diffusion-aware operators. Standard operators are uniform crossover and bit flip mutation, while proposed operators are Phase Aware Crossover and Block Mutation.

Method	Fitness ↑	Convergence
Simple GA + Standard Ops	92.45	Slow
Island Model + Standard Ops	94.50	Medium
Simple GA + Proposed Ops	95.25	Medium
Island Model + Proposed Ops	96.88	Fast

Table 3

Runtime and memory comparison

Method	Time (s) ↓	VRAM (GB) ↓
Stable Diffusion	1.00	14.2
DeepCache	0.72	12.5
Ours	0.48	10.8

$$\mathbf{x}'_{t:t+k} = 1 - \mathbf{x}_{t:t+k} \quad (13)$$

where k is the block size sampled from a small range as shown in fig 3. This enables structures exploration while maintaining temporal smoothness. The mutation probability is applied at block level:

$$\mathbf{x}' = \begin{cases} \text{BlockFlip}(\mathbf{x}) & \text{with probability } p_m \\ \mathbf{x} & \text{otherwise} \end{cases} \quad (14)$$

3.3.3. Fitness Function

We define a multi objective fitness function that balances image quality and efficiency.

$$F(\mathbf{S}) = w_{ssim} \cdot \text{SSIM}(\mathbf{I}_{base}, \mathbf{I}_{dist}) + w_{psnr} \cdot \text{PSNR}(\mathbf{I}_{base}, \mathbf{I}_{dist}) - w_{eval} \cdot \frac{|\mathbf{S}|}{T} \quad (15)$$

where $\mathbf{I}_{base}$ is the baseline image, $\mathbf{I}_{dist}$ is the generated image using cache schedule $\mathbf{S}$ and $\frac{|\mathbf{S}|}{T}$ is the evaluation ratio.

We use SSIM and PSNR in fitness function because they are computationally cheap and fast to calculate on-the-fly, allowing the Genetic Algorithm to rapidly evaluate hundreds of candidate schedules in minutes. However, we ensure the validity of these schedules by evaluating the final optimized results using robust perceptual metrics (FID and CLIP) in table 1. Because the Genesis framework is completely modular, practitioners who prioritize human perceptual alignment over search speed can easily swap SSIM/PSNR for learned perceptual metrics like LPIPS.

3.3.4. Migration Strategy

To prevent stagnation, we implemented periodic migration[Shapiro (1999)] using a ring topology:

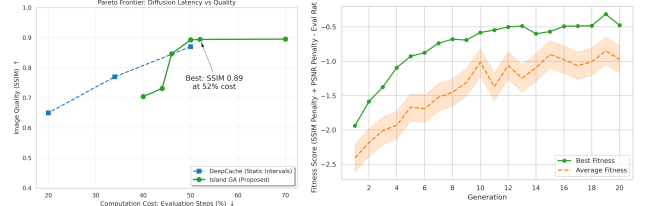


Figure 4: (Left) Trade-off between inference efficiency and image quality. Our method achieves higher SSIM[Bakurov et al. (2022)] at lower evaluation step ratios compared to DeepCache(Uniform), demonstrating a better quality and efficiency balance. (Right) Evolution of fitness across generations. The best fitness steadily improves while the average fitness increases with reduced variance, indicating stable convergence of the optimization process.

$$\mathcal{P}_i \rightarrow \mathcal{P}_{(i+1) \bmod N} \quad (16)$$

During migration, top k individuals from each island replace the worst performing individuals in the neighboring island:

$$\mathcal{E}_i = \text{Top-}k(\mathcal{P}_i) \quad (17)$$

$$\mathcal{P}_{i+1}^{worst} \leftarrow \mathcal{E}_i \quad (18)$$

4. Experiments

4.1. Setup

All experiments were conducted on NVIDIA H200 GPUs with 141GB HBM3 memory using CUDA 12.4. We evaluated our approach on two datasets: MS-COCO and PartiPrompt. We also profile inference runtime and peak GPU memory using PyTorch[Imambi, Prakash and Kanagachidambaresan (2021)] CUDA profiling tools to evaluate computational efficiency. We used Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5), PNDMScheduler, 50 denoising steps with image resolution of 512×512. We evaluated 5000 prompts per dataset with seed 42 to ensure reproducibility. The population size is 10 and number of islands were 5 with 20 generations. The fitness weight values are taken as $w_{ssim} = 100$, $w_{psnr} = 1$, $w_{eval} = 10$.

4.2. Evaluation Protocols

Our evaluation is divided into two stages: optimization validation and generative performance evaluation.

4.2.1. Optimization Benchmarks

To validate the robustness of the optimization strategy, we evaluated the proposed method on benchmark functions:

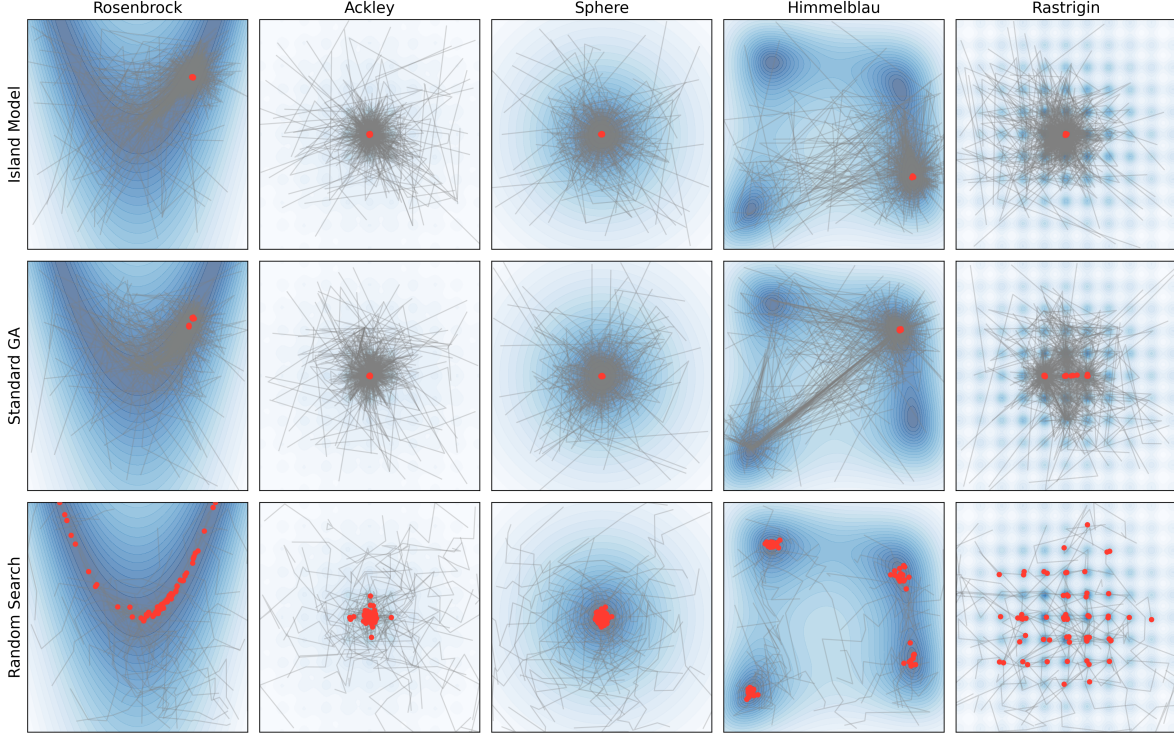


Figure 5: Optimization trajectories of three search strategies Island Model (top row), Standard Genetic Algorithm (middle row), and Random Search (bottom row) across five benchmark functions: Rosenbrock, Ackley, Sphere, Himmelblau, and Rastrigin. Background contours represent the objective landscapes, gray lines denote search trajectories, and red markers indicate sampled candidate solutions, with final/high-quality solutions highlighted. The Island Model demonstrates more stable and directed convergence toward global optima, while Standard GA exhibits moderate exploration with occasional convergence, and Random Search shows scattered sampling with limited convergence efficiency, particularly on multimodal landscapes (Ackley, Rastrigin, Himmelblau).

Rosenbrock Function [Jarre (2013)]

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2 \quad (19)$$

Rastrigin Function [Beyzade, Atan and Saraçoğlu (2019)]

$$f(x) = 10n + \sum_{i=1}^n [x_i^2 - 10 \cos(2\pi x_i)] \quad (20)$$

Additional benchmarks include Ackley[Ackley (2014)], Himmelblau[Paviani and Himmelblau (1969)] Functions. These experiments demonstrate the global search capability and convergence stability of the proposed method as shown in fig 5.

The cache schedule comparison (Fig. 7) shows how our method selects a sparse and adaptive set of evaluation steps, in contrast to the uniform intervals used by DeepCache. This adaptive selection reduces unnecessary full evaluations while preserving image quality.

Fig. 4 (left) shows task specific Pareto frontiers for MS-COCO & PartiPrompt datasets. Each curve represents the trade off between inference efficiency & image quality. We observe that Pareto frontier shifts depending on dataset. For MS-COCO the curve is smoother which indicates stable quality under aggressive caching and PartiPrompt shows sharper degradation due to high semantic diversity.

Fig. 4 (right) shows the fitness evolution over generations. The best fitness curve rises steadily while variance in the average fitness decreases, confirming that the Island Model converges stably without premature collapse.

4.2.2. Runtime and Memory Profiling

To ensure practical relevance we evaluate computational efficiency in terms of runtime and peak memory usage. Our method achieves less runtime & memory usage due to reduced full model evaluations & optimal caching schedules as shown in table 3.

4.3. Ablation Study

Table 4 presents an ablation analysis of different search strategies within the proposed optimization framework. Starting from a Random Search, the introduction of a Standard Genetic Algorithm leads to a significant improvement in fitness, increasing from 84.102 to 92.4501 and reducing the optimization error across all benchmark functions.

Building upon this, the proposed Island Model further enhances performance by incorporating multiple sub-populations and periodic migration. This results in the highest fitness score (96.882) and the lowest error values across all evaluated functions including Sphere, Ackley and Rastrigin. Consistent improvements demonstrate the advantage of maintaining population diversity.

To evaluate the impact of our proposed operators we compare the Island Model with standard crossover and bit flip mutation against our proposed operators in table 2. The results show that using Phase Aware Crossover and Block Mutation leads to faster convergence and higher fitness.

5. Results

The experimental results demonstrate that the proposed Genesis framework consistently improves the efficiency-quality trade-off for diffusion model inference.



Figure 6: Qualitative comparison across diffusion models and caching strategies. Images generated from multiple text prompts are shown for Stable Diffusion[Rombach et al. (2022)], BK-SDM Base[Kim et al. (2024)], Small SD[Kim et al. (2024)], DeepCache[Ma et al. (2024)], CacheQuant[Liu et al. (2025)] and our method.

5.1. Quantitative Results

Table 1 shows that Genesis reduces the computational cost from 133.58T BOPs (DeepCache) to 80.00T BOPs corresponding to approximately 40.1% fewer operations while simultaneously improving image quality. On MS-COCO, Genesis achieves the lowest FID of 20.32 outperforming DeepCache (23.45), TokenCache (23.80) and CacheQuant (23.55). Similar improvements are observed on PartiPrompts where Genesis achieves the highest CLIPScore (27.12), PSNR (32.85) and SSIM (0.855) indicating superior semantic alignment. These improvements demonstrate that optimizing the cache schedule is more effective than uniformly skipping the evaluation steps.

5.2. Qualitative Results

Figure 6 presents a qualitative comparison between Stable Diffusion, BK-SDM, Small SD, DeepCache, CacheQuant and the proposed Genesis framework across diverse text

prompts. The generated images demonstrate that Genesis preserves both structural fidelity and semantic consistency while substantially reducing inference time. For prompts requiring strong global structure, such as "the silhouette of an elephant on the full moon" and "a black and white landscape photograph of a black tree," Genesis maintains well-defined object boundaries, accurate scene composition, and realistic lighting comparable to the full Stable Diffusion baseline. Unlike lightweight diffusion variants, which occasionally simplify object geometry or lose fine structural details, Genesis produces images with higher visual fidelity despite requiring significantly fewer evaluation steps.

6. Conclusion

In this paper, we proposed an evolutionary framework to overcome the limitation of using uniform cache schedules. We reformulate the selection of the caching schedule as an optimization problem. Using the Island Model, our approach

Table 4

Performance comparison of search techniques on fitness and benchmark optimization tasks. Higher fitness values indicate better solutions, while lower error values correspond to improved optimization performance.

Task		Random Search	Standard GA	Island Model
Fitness		84.102	92.4501	96.882
Error	Sphere	12.45	0.05	0.0014
	Ackley	19.12	3.02	0.84
	Rastrigin	145.20	45.21	12.10

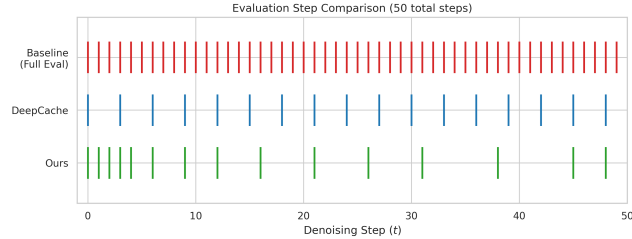


Figure 7: Comparison of evaluation schedules across 50 denoising steps. The baseline performs full evaluation at every step while DeepCache reduces evaluations via uniform caching. Our method further sparsifies evaluations through adaptive step selection achieving more efficient computation with fewer evaluation points.

searches for the optimal cache schedule over generations. Experiments validate both the robustness of the optimization process and effectiveness of the cache schedule. Evaluation on standard optimization benchmarks demonstrates stable convergence and global search capability. Experiments on MS-COCO and PartiPrompt show that the optimized schedule maintains high visual quality and semantic alignment.

7. Limitations

- Genesis successfully identifies optimal cache schedules but the evolutionary search process introduces one time computational overhead prior to deployment.
- The default configuration uses a population size of 10 across 5 islands, evolving over 20 generations. This results in exactly 1000 ($10 \times 5 \times 20$) schedule evaluations.
- This is a one-time offline search. Since the discovered optimal schedule is static during deployment, the one time search cost results in zero overhead at runtime.

8. Acknowledgement

The authors acknowledge the funding support from ANRF, India under the PMECRG scheme (IITM/ANRF/KP/529/2025, ANRF/ECRG/2024/004417/ENS) and scholarship support from MHRD, India.

9. Declaration of Generative AI use

During the preparation of this work the author(s) used ChatGPT in order to paraphrase the content for better understanding. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

- Ackley, D.H., 2014. A connectionist algorithm for genetic search¹, in: Proceedings of the First International Conference on Genetic Algorithms and their Applications, Psychology Press. p. 121.
- Ao, P., 2005. Laws in darwinian evolutionary theory. *Physics of life Reviews* 2, 117–156.

- Bakurov, I., Buzzelli, M., Schettini, R., Castelli, M., Vanneschi, L., 2022. Structural similarity index (ssim) revisited: A data-driven approach. *Expert Systems with Applications* 189, 116087.
- Baldwin, J.M., 2018. A new factor in evolution. *Diacronia*, 1–13.
- Beyzade, S., Atan, Ö., Saraçoglu, R., 2019. Finding the best population size for optimizing the rastrigin function using genetic algorithm, in: 4th INTERNATIONAL CONFERENCE ON ADVANCES IN NATURAL & APPLIED SCIENCES, p. 457.
- Darwin, C., 2016. On the origin of species, 1859.
- Gardner, A., Welch, J.J., 2011. A formal theory of the selfish gene. *Journal of Evolutionary Biology* 24, 1801–1813.
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y., 2020. Generative adversarial networks. *Communications of the ACM* 63, 139–144.
- Ibtehaz, N., Kihara, D., 2023. Acc-unet: A completely convolutional unet model for the 2020s, in: International conference on medical image computing and computer-assisted intervention, Springer. pp. 692–702.
- Imambi, S., Prakash, K.B., Kanagachidambaresan, G., 2021. Pytorch, in: Programming with TensorFlow: solution for edge computing applications. Springer, pp. 87–104.
- Jarre, F., 2013. On nesterov’s smooth chebyshev–rosenbrock function. *Optimization Methods and Software* 28, 478–484.
- Kim, B.K., Song, H.K., Castells, T., Choi, S., 2024. Bk-sdm: A lightweight, fast, and cheap version of stable diffusion, in: European Conference on Computer Vision, Springer. pp. 381–399.
- Lin, T.Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L., 2014. Microsoft coco: Common objects in context, in: European conference on computer vision, Springer. pp. 740–755.
- Liu, X., Li, Z., Gu, Q., 2025. Cachequant: Comprehensively accelerated diffusion models, in: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 23269–23280.
- Ma, X., Fang, G., Wang, X., 2024. Deepcache: Accelerating diffusion models for free, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 15762–15772.
- Paviani, D., Himmelblau, D.M., 1969. Constrained nonlinear optimization by heuristic programming. *Operations Research* 17, 872–882.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B., 2022. High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 10684–10695.
- Shapiro, J., 1999. Genetic algorithms in machine learning, in: Advanced course on artificial intelligence. Springer, pp. 146–168.
- Tanchenko, A., 2014. Visual-psnr measure of image quality. *Journal of Visual Communication and Image Representation* 25, 874–878.
- Weng, L., 2021. What are diffusion models? lilianweng. github. io, 21.
- Whitley, D., Rana, S., Heckendorn, R.B., 1999. The island model genetic algorithm: On separability, population size and convergence. *Journal of computing and information technology* 7, 33–47.
- Yu, J., Xu, Y., Koh, J.Y., Luong, T., Baid, G., Wang, Z., Vasudevan, V., Ku, A., Yang, Y., Ayan, B.K., et al., 2022. Scaling autoregressive models for content-rich text-to-image generation. *arXiv preprint arXiv:2206.10789* 2, 5.
- Zhang, Y., Hartl, B., Hazan, H., Levin, M., 2024. Diffusion models are evolutionary algorithms. *arXiv preprint arXiv:2410.02543*.

Highlights

Manuscript Title:

Genesis: Evolving Optimal Cache Schedule for Diffusion Models

- We reformulate Cache scheduling as a structured evolutionary optimization problem.
- We introduce diffusion aware operators Phase Aware Crossover and Block Mutation.
- Genesis reduces the computational cost while simultaneously improving image quality.
- On MS-COCO, Genesis achieves the FID 20.32 outperforming other methods.
- On PartiPrompts, Genesis achieves the CLIPScore(27.12), PSNR(32.85) & SSIM(0.855).

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: